



Optimization of CNN Using Transfer Learning

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MENTOR

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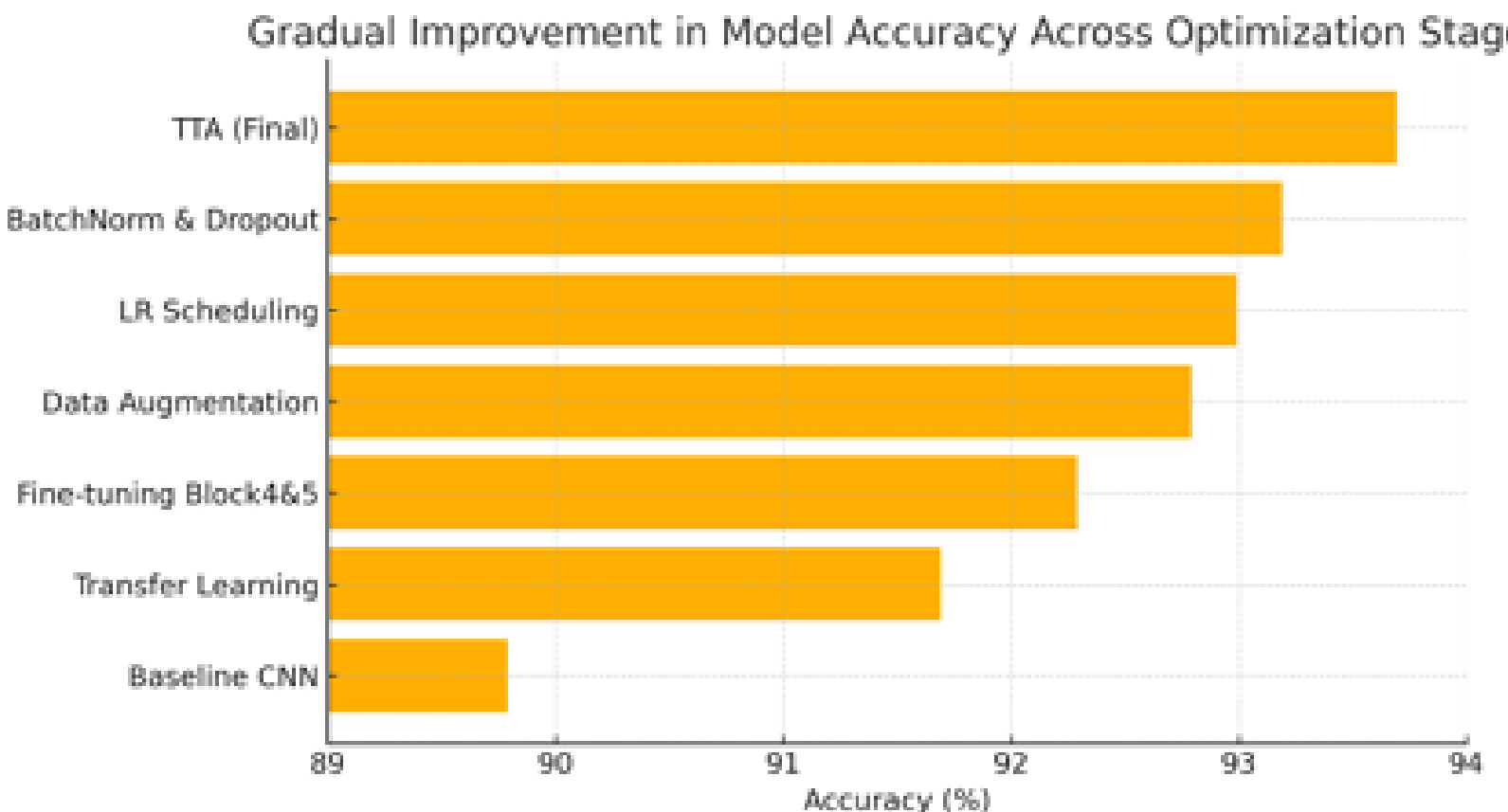
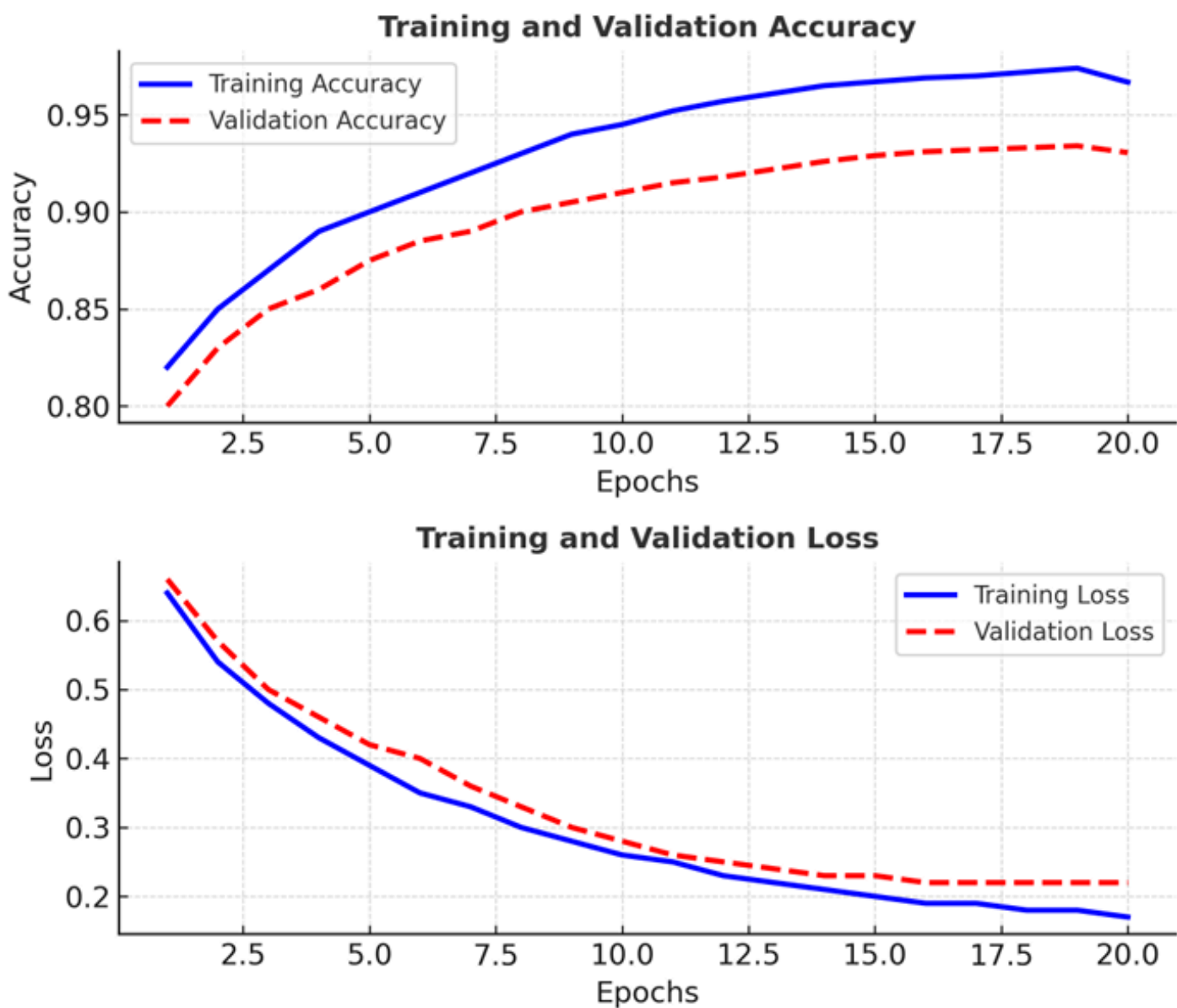
INTRODUCTION

Deep learning models like VGG16 achieve high accuracy but need heavy computation. This work uses transfer learning with an optimized VGG16-CNN to ensure stable, deterministic inference under limited GPU memory—ideal for real-time, embedded use.

METHODOLOGY

VGG16 is fine-tuned on Fashion-MNIST with data augmentation (flips, rotations, brightness). A lightweight CNN head with BatchNorm, Dropout, Softmax replaces dense layers. Training uses Adam, LR scheduling, and 2 GB GPU capping for consistent performance.

RESULT



FUTURE SCOPE

Extend with quantization and light models like MobileNet and EfficientNet-Lite for faster, more efficient deployment.

CONCLUSION

The optimized VGG16-based model achieves 93.69% accuracy with stable, low-latency inference (≈ 21 ms) under a 2 GB VRAM limit. It ensures reproducibility, speed, and efficiency, making it suitable for real-time deployment on resource-constrained platforms.